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Optional Introduction to Jupyter and R

Introduction to Generalized Linear Models

Binomial Regression

Binomial Regression and Goodness of Fit

Ethical Issues in Statistics and Data Science

Reading: Fair ML Book, Introduction
10 min

Peer-graded Assignment: Ethical Issues in Statistics and Data Science (Fair ML Intro)
Grading in progress

Review Your Peers' Ethical Issues in Statistics and Data Science (Fair ML Intro)

Assignments: GLMs for Binomial Data

Peer-graded Assignment: Ethical Issues in Statistics and Data Science (Fair ML Intro)

Reviews 3 left to complete

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W1 PR 1

by Anonymous Learner
August 21, 2024

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PROMPT How are the models that we've used in this course so far (and in this specialization more broadly) based on the process of induction? The models used in the specialisation so far is based on the process of induction in that, based on examples, it generalises. The linear models used throughout the last course in the specialisation used the training samples to tune the parameters of the model so that we were able to determine an approximate value for an unknown sample based on it's similarity to the samples the model was trained on.	RUBRIC Did the student correctly identify how models used in this course are based on induction? ☐ 0 pts No Answer ☐ 4 pts The student attempted to answer the question, but the response does not demonstrate an correct understanding of induction. ☐ 6 pts The student correctly identified that our models are based on induction, but the answer contained some moderate errors. ☒ 8 pts The student correctly identified that our models are based on induction, but the answer contained some minor errors. ☐ 10 pts Yes, the student correctly identified that our models are based on induction, and gave specific examples and explanations.
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PROMPT "The fact that machine learning is 'evidence-based" by no means ensures that it will lead to accurate, reliable, or fair decisions." Provide 2-3 examples of this claim. The fact that machine learning is 'evidence-based" by no means ensures that it will lead to accurate, reliable, or fair decisions because: 1) The data itself might be biased or the interpretation of it might introduce bias. A concrete example could be a model based on arrest records. Seems neutral and natural to conclude that if a certain demographic is over-represented in that data then it is because they are more likely to commit crime. However, it could be that the police is more active in certain areas than others, increasing the risk of being arrested in those areas compared to other areas which might lead to a misleading picture. It could also be that certain demographic groups could be at a higher risk of getting arrested but not convicted. All of this could lead to an unfair model even if the model itself is technically and mathematically correct. 2) If certain demographics are under-represented in the training data (or over-represented) this could lead to a biased model as well. A well-known example is lack of people with dark skin in certain datasets used to train image recognition models leading to models that make wrong categorisations. 3) Even if the model is neutral in judgements it might not be fair. Sometimes a biased model might be preferable. One example, also mentioned in the text, could be a model that judges whether or not a convicted person is likely to commit crime again. It might be that young people are more likely to commit crime again but on the other hand being lenient might lead to the young person escape from a life in crime.	RUBRIC Did the student provide 2-3 examples? ☐ 0 pts No answer ☐ 3 pts The student provided inadequate examples, or only one example. ☐ 6 pts The student provided 2-3 examples.
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PROMPT Do you believe that Amazon's same-day delivery system mention on page 3-4 is unfair or unjust? Why or why not? Personally I think it might be fair. Without knowing the details of US postage rates or the inner workings of Amazon's model I find it difficult to answer that question. If the price is indeed higher for Amazon to offer same-day delivery to some neighbourhoods than others then Amazon is just reflecting the situation. Would it be fair to ask a private company to try to solve a deeper underlying problem, potentially losing money in the process? In reality that additional cost would end up with the customers anyway, eventually. Averaging out the cost would also be problematic; would it not be equally unfair to those living in the "better" neighbourhoods to start charging them more? To me it sounds like a problem that needs to be addressed(and is being addressed) by the government rather than by Amazon.	RUBRIC Did the student answer the question and provide a justification for their answer? ☐ 0 pts No answer ☐ 4 pts The student answered the question but their justification had major gaps in reasoning. ☐ 6 pts The student answered the question but their justification had some gaps in reasoning. ☐ 8 pts The student answered the question and provided an adequate justification for their answer.
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PROMPT What is the machine learning loop? Do you think that the machine learning loop, given in figure 1, applies to statistical modeling? Justify your answer. The machine learning loop refers to the cycle where the state of the world is measured, those measurements then turned to data that is used to train models and the predictions from those models then affecting individuals, thereby changing the state of the world. That loop is a continuing process and there is some interpretation and risk of errors in each of those steps that can influence the world around us in a positive but certainly also a negative direction.	RUBRIC Did the student state what the machine learning loop is, and state and justify why they think it does (or does not) apply to statistical modeling? ☐ 0 pts No answer ☐ 4 pts The student correctly stated what the machine learning loop is, but did not state why they think it does (or does not) apply to statistical modeling. ☐ 7 pts The student correctly stated what the machine learning loop is, and stated why they think it does (or does not) apply to statistical modeling, but the justification was lacking. ☐ 10 pts The student correctly stated what the machine learning loop is, and stated and justified why they think it does (or does not) apply to statistical modeling.
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